ZOO-514 OBJECTIVES

MID TERM- 100%

PRINCIPLES OF ANIMAL LIFE-I

GUESS PAPER 2023

BY SULMAN ALI GUJJAR

0341-7547044

OBJECTIVES

1. Lipids are soluble in ..... ***Organic solvent***
2. Lipids are polar or Non- polar?***Non-polar***
3. Lipids are made up of which compounds? ***C,H and O***
4. Write the name of some important types of lipids..... ***Triglycerides, Phospholipids, Steroids***
5. Name five sences of observation , ***Smell, Taste, Touch , Hearing, Vision***
6. The information obtained from observation is called..... ***Data***
7. How many types of data, ***Two***
8. Name two types of data ***Qualitative observation, Quantitative observation***
9. Resources that are exist naturally called…***Natural Resources***
10. In which we study in zoology.... ***study of animal life***
11. ***Anatomy:*** It is the study of the structure of complete organisms and their body parts.
12. ***Lipid:*** A diverse group of biomolecules which are water insoluble, Non-polar, Readily soluble in organic solvent.
13. Amino acid have which type of bond… ***peptide bond***
14. ***Histology:*** Study of organisms at tissue level is called histology.
15. Almost all forms of life have same ***genetic blue prints (DNA).***
16. Lipids are polar or Non- polar?...***Non-polar***
17. ***Aquaculture*** is the culturing of aquatic animals and plants.
18. ***Anatomy:*** It is the study of the structure of complete organisms and their body parts.
19. ***Physiology:*** The study of the function of organisms and their body parts.
20. Evolutionary theory is also based on the idea that all organisms evolved from a ***common ancestor.***
21. The branch of science that deals with the naming and classification of organisms into different groups and subgroups on the basis of relatedness is called ***taxonomy.***
22. The major criteria of theory are: ***Observable, repeatable and predictive/ Explanatory.***
23. ***Monosaccharides: "***The simplest carbohydrates which cannot be further hydrolyzed into simpler carbohydrates"

24. ***Glucose*** is the most common type of ***monosaccharides***

25. ***Disaccharides*** "Two monosaccharides combine together to form a disaccharide"

26. ***Cell*** is the structural and functional unit of life.

27. Lipids / Fatty acids Having single covalent bond between all carbons are ***saturated.***

28. Lipids / Fatty acids Having single covalent bond between all carbons are ***saturated.***

29. Atoms in biomolecules are held together mainly by ***covalent interaction***

30. Example of Proteins with Tertiary Structure ***Myoglobin, Salivary amylase***

31. Many ***animals*** play their role in cleaning the environment.

32. This earth are at the risk of extinction because of ***overexploitation*** and destruction of their natural habitat.

33. Melting point of Palmitic acid… ***62.9***

34. ***Cell biology (Cytology):*** Study of the structure and function of cell and its components.

35. ***Molecular Biology:*** Molecular biology is the study of structure and function of the macromolecules (e.g. proteins and nucleic acids, Lipids) essential to life.

36. ***Entomology:*** Entomology is a branch of zoology which deals with the study of insects.

37. ***Malacology:*** Malacology is the branch of invertebrate zoology that deals with the study of the mollusca.

38. ***Ichthyology:*** Ichthyology is the branch of zoology concerned with the study of fishes.

39. ***Herpetology:*** Herpetology is the branch of zoology that deals with the study of amphibians and reptiles.

40. ***Ornithology:*** It is the branch of zoology concerned with the study of birds.

41. ***Histology:*** Study of organisms at tissue level is called histology.

42. ***Embryology:*** Study of development of embryo from fertilized egg.

43. ***Genetics:*** Study of the mechanisms of transmission of traits from parents to offspring.

44. ***Protozoology:*** Protozoology is the study of protozoa. It deals with the characterization, taxonomy and medical importance of protozoa.

45. ***Nematology:*** Nematology is the scientific discipline concerned with the study of nematodes, or roundworms.

46. ***Mammalogy:*** It is the branch of zoology that deals with the study of mammals.

47. ***Macromolecules*** are (e.g. proteins and nucleic acids, Lipids)

48. Beautiful domestic pets, wildlife, stunning aquaculture are all studied under the heading of ***zoology.***

49. Overall folding of polypeptide chain into a 3D form is called ***tertiary structure.***

50. Example of Proteins with Tertiary Structure , ***Myoglobin Salivary amylase***

51. ***Animal diversity:*** Animal diversity is the number of different types of animals living on this earth.

52. They may be used Insects to control the pest. The phenomenon is called ***biocontrol.***

53. ***Animal husbandry*** is the farming of animals to produce meat, milk and eggs

54. ***Aquaculture*** is the culturing of aquatic animals and plants.

55. Beekeeping for the production of honey and wax is called ***apiculture***

56. Rearing of silkworm for production of silk is called ***sericulture.***

57. The branch of biology which deals with the study of animals life is called? ***Zoology***

58. Gr. zoon mean ..... and logos means..... ***Animlas, study***

59. Two monosaccharides combine together to form? ***Disaccharides***

60. Closely related species form….a ***Genus***

61. ***Proteins:*** Biomolecules having one or more chains of amino acids.

62. ***Polysaccharides:*** "When many monosaccharides join together through glyosidic bond, they form a polysaccharide"
63. ***Theory:*** When a hypothesis confirmed by repeated experiments and other related hypothesis support then it is converted to a theory.
64. ***Order:*** Closely related families form an order. e.g., Carnivora and Primates
65. ***Hemoglobin*** carries oxygen from lungs to capillaries
66. Comparison of hemoglobin structure of human and fish is example of ***Evolutionary science***
67. ***Invertebrates are*****:** Echinoderms , Arthropods , Mollusca~, Annelida , Nematodes, Platyhelminthes , Cnidarian, Porifera
68. ***cAMP*** work as chemical messenger within the cell
69. ***Aedes*** mosquitos are responsible for spread of dengue.
70. Write some essence of zoology? ***Fasnitating animals life, Domasticated pets to wildlife, Beautiful aquatic animlas***
71. Muslims scientist give many descovries in the field of ..... ***Zoology***
72. Modern scientists revolutionized the science of..... ***Zoology***
73. The process of identifying a disease using signs and symptoms is called.....***Disease diagnoses***
74. ..... is the study of uses, effects, and mode of action of drugs is called? ***Pharmacology***
75. ..... is farming of animals to produced foods such as meat, milk and eggs ***Animal Husbandry***
76. ..... is the rearing of aquatic animals or the cultivation of aquatic plants for food ***Aquaculture***
77. ..... is the maintanance of bee colonies, commomly in made man hives. ***Apiculture***
78. Silk production from silk worms is called..... ***Sericulture***
79. DNA has same .....composition ***Chemical***
80. .....carries genetic information ***DNA***
81. ..... is the fundamental unit of lufe ***Cell***
82. .....makes all the structures of living organisms ***Cell***
83. Name some common evolutionary forces ***Genetic drift, Mutations, Gene flow, Inbreeding, Selection***
84. All animals get food from..... ***Environment***
85. ***Anotomy*** ; Study of structure and complete organisms and part of thier body.
86. ***Physiology***; Study of the function of organisms \and thier part.
87. ***Cell biology***; Study of the structure and function of cell and thier components.
88. ***Molecular biology;*** Study of the structure and function of the macromolecules (e.g. Protiens lipids and nucliec acid) essiential for life.
89. ***Histology***; Study of functionof organisms
90. ***Genetics***; Study of the machanisms of transmission of traits from parents to offsprings
91. Specialization of zoology on the basis of animal group.....***Prozoology***;
92. Study of protozoa ***nematodology***; Study of nematodes or roundworms.
93. ***entomology***;Study of insects
94. ***malacology***;Study of mollucus
95. ***Ichthyology***; Study of Fish
96. ***Herpetology***;Study of amphibians and reptile
97. ***Ornithology***; Study of birds

98. ***Mammalogy***;Study of mammals

99. The study of protozoa is called.....***Protozology***

100. Nematology is the scientific discipline concerned with the study of

101. ..... or .....? ***Nematodes, earthworms***

102. ..... is the branch of zoology which deal with the study of insects. ***Entomology***

103. Malacology is the branch of zoology which deals with the study of ..... ***Mulluscus***

104. The branch of zoology which deals with the study of fishes is called.....***Ichthyology***

105. ..... is the branch of biology which deals with the study of amphibians and reptiiles. ***Herpetology***

106. The study of birds is called.....***Ornithology***

107. Mammalogy is the study of ..... ***Mammals***

108. The number of different types of animals living on this earth is called ..... of animals. ***Biodiversity***

109. Estimates range of species on earth is..... million ***30***

110. How many species have been descoverd? ***80%***

111. How many species have been extinct? ***>99%***

112. The branch of science which deals with the naming and calssification the organisim is called..... ***Taxonomy***

113. Arrangment of animals into different groups and sub groups on the basis of relatedness is called..... ***Classificaion of animals***

114. .....is a group of animals which can inbreed and produced fertile offsprings. ***Species***

115. Dogs, cats ,and elephants are the examples of ..... ***Spieces***

116. Panthera is a genus of .....***wild cats***

117. Hominidae and Felidae are ..... ***Family***

118. Cornivora and Primates are..... ***Order***

119. Amphibia,Reptilia, Aves, Mammalia are..... ***Class***

120. Classes with some common features are grouped to form..... ***Phylum***

121. Chordata , Echinodermata are ..... ***Phylum***

122. Name five Kingdomes***Monera protesta fungai plantie animalia***

123. Life is divided into how many domains? ***Three***

124. Name three domains of life..... ***Archaea Becteria, Eukarya***

125. Any kind of stock or material that can be used for working effectively is called..... ***Resource***

126. .....resource are exist without the envolvement of mankind. ***Natural resource***

127. Magnetic, Gravitational force, Sunlight, Atmosphere, plants animals are the example of .....resource ***Natural resource***

128. How many types of natural resource? ***Two***

129. Name two types of natural resource..... ***Renewable resource, Non-renewable resource***

130. .....resources are resonably replace or recovered. ***Renewable resource***

131. Water, sunlight, air, biomass are ..... resource***Renewable resource***

132. ..... resources are not replaces or recovered. ***Non-renewable resource***

133. Oil, gas, menirals are..... resource ***Non-renewable resource***

134. What are the threats to resources ***Over exploitation Miniral reserves are limited Oil reservoir limited loss of biodiversity pollution***

135. What is the solution to save our resources? ***Conservation, Alernative resource, Recycling, Pollution control***

136. Current population of the world is ..... ***7.7 billion***

137. The number of people in less developed countries will..... in future ***Increase***

138. Current population growth rate is..... its means .....per years ***1.07%, 82 billions***

139. In 1960 growth rate was..... ***2%***

140. A series of steps followed by scientists to answer a specific natural problem is called ..... ***scientific mathod***

141. Collecting information about soecific problems is called ..... ***Observation***

142. How many sences of observation ***5 sences***

143. Biomolecules are chemically cinsists of..... ***Oxygen, Carbon, Hydrogen ,Nitrogen,Sulfur, Phosphorus***

144. Atoms in biomolecules are held together by..... ***Covelent intraction***

145. Basic building blocks of living organisms are ..... ***Biomolecules***

146. How many types of biomolecules? ***Four***

147. Name four types of biomolecules.....***protiens, lipids, carbohydrates, and nucliec acid***

148. Most abundant biomolecule on eart are..... ***Carbohydrate***

149. .....are polyhydroxi aldehydes ketones or complex molecules giving aldihyde or ketones on hydrolysis. ***Carbohydrate***

150. Carbohydrates are made up of..... ***C,H, and O***

151. General formula of carbohydrates are..... ***$(CH_2O)n$***

152. In how many classes carbohydrates are classified? ***Three classes***

153. Name the classes of carbohydrates ***Monoseccharides Diseccharides/ oligoseccharides Polysaccharides***

154. The simplest carbohydrates which cannot be furher classified are called..... ***Monosaccharides***

155. Eaxmples of monosaccharides..... ***Glocuse***

156. Monosccharides are ..... in taste ***Sweet***

157. Two monosaccharides combine to form ..... ***Disaccharides***

158. Write general formula of Diascharides..... ***Cx(H2O)y***

159. Write some common Diasaccharides..... ***Sucrose, Lactose, Maltose***

160. Molecules with same molecular formula but different structural formula are called..... ***Isomerisms***

161. Write the molecular formula of isomerism is..... ***C12H22O11***

162. ..... bond join monosaccharides subunit ***Glycosidic bond***

163. When many monosaccharides join together through glycosidic bond , they form..... ***Polysaccharide***

164. General formula of polysaccahrides is..... ***Cx(H2O)y and (C6H10O5)n***

165. Write name of som polysaccharides. ***Glycogen, Starch, Cellulose, Chitin***

166. Glycogen and starch play role in ..... ***Fuel storage***

167. Cellulose and chitin plays role in ..... ***Structural role***

168. Carbohydrates provides .....% energy. ***64-65%***

169. How many amounts of energy carbohydrates provide per day? ***225-325 Grams per day***

170. ..... is primarily synthesized by plants during photosynthesis. ***Glucose***

171. ..... is more important source of energy for all organism ***Monosacchrides***

172. We get sucrose from..... ***Sugarcane and Sugarbeet***

173. Luctose present in ..... ***Milk sugar***

174. Maltose is present in ..... and ..... ***Sweet potatoes and candies***

175. Sucrose, Lactose , and Maltose are..... ***Diasccharides***

176. Starch, Glycogen and cellulose are ..... ***Polysacharides***

177. ..... is digested into glocuse. ***Starch***

178. ..... quickly breaks to provides glocuse into cells and blood. ***Glycogen***

179. ..... breaks down into glocuse. ***Cellulose***

180. Exoskeleton of different of different arthopodes such as insects , crustacean are made up of..... ***Chitin***

181. .....provide to cell wall of plants. ***Cellulose***

182. Extracellular material is present in ..... ***Fluid of joints, Vitreous humor, Tendons, Lagaments, Skin***

183. A diverse group of water molecules which are insoluble in water is called....., ***Lipids***

184. Examples of Quantitaive data, ***Height, Weight, Temperature***

185. Examples of Qualitative data ***Shape, Colour***

186. .....use your senses to observe result is called, ***Qualitative data***

187. .....use instruments such as rulers, balances, beakers, thermometer etc. ***Quantitative observation***

188. A reasonable explaination to the natural question is called..... ***Hypothesis***

189. How many types of hypothesis? ***Two***

190. Name two types of hypothesis. ***Deductive reasoning, Inductive reasoning***

191. Parameters in experiment that may change is called..... ***Variables***

192. How many types of variables?, ***Three***

193. Name three types of variables.... ***Independent variables, Dependent variables, Controlled variables***

194. A..... is based on repeated experimental observations that describes some aspect of nature. ***Scientific law***

195. Examples of scientific law ***Law of segregation, Law of independent assortment, Hardy-Weinberg law***

196. Peroximate causes also called, ***Immediate causes***

197. Immediate repones of causing is called..... ***Peroximate causing***

198. Write some examples of peroximate causing... Matabolic and physiological functions

199. Ultimate causing are also called..... ***Real causing***

200. ..... which produced systems and its characteristic through evolutionary time. ***Ultimate causing***

201. Write example of ultimate causing.....***Evolutionary forces that cause birds to aquire migratory behavior.***

202. Biological sciences that adress the questions of peroximate causes is called..... and it proceeed through experiments. ***Experimental sciences***

203. Write some examples of experimental sciences. ***Process of maintaining body temperature, process of cell division.***

204. Biological sciences that address the questions to ultimate causes is called..... and it proceed through comparative methods. ***Evolutionary sciences.***

205. Write some examples of evolutionary sciences. ***Camparison of hemolglobin structure of human and fish.***

206. A species of atom having the same number of protons in thier atomic nuclie is called..... ***Element***

207. How many elements make living organisim? ***25 elements***

208. ..... are formed by atoms of two or more elements . ***Compounds***

209. Write some compounds name..... ***Fatty acid,Amino acids, water , Glocouse,Nucleotides. etc.***

210. Atoms and molecules are held together by..... ***Chemical bonding***

211. How many types of bond? ***Three***

212. Name three types of bond...***Covelent bond, Hydrogen bond, Ionic bond***

213. Acids release ..... ions. ***H ion***

214. Write some acids name..... ***Nucliec acid, Hydrochloric acid, Fatty acid, Amino acid***

215. Bases release .....ions. ***OH ions***

216. Write some bases name..... ***Nitrogen base***

217. Buffers are with ..... pH. ***Stablize***

218. The stablize pH of buffer is..... ***(7.35-7.45)***

219. Too much fluctuation in pH of buffer can.....cell ***distroy***

220. Molecules produced by cells or living organisms is called..... ***Biomolecules***

221. How many types of biomolecules? ***Two***

222. Name two types of biomolecules..... ***Macmolecules, Micromolecules***

223. Molecules with large molecular weight are called..... ***Macromolecules***

224. Molecules with low molecular weight are called..... ***Micromolecules***

225. Write some exanples of micromolecules..... ***Protiens, lipids and starch etc***

226. write some examples of macromolecules..... ***Monosccharides, Amino acids etc***